# Least common denominator

**1** Find the least common denominator of the following pairs of algebraic fractions:

**a** $\frac{s}{3}$ and $\frac{s}{9}$

**b** $\frac{a}{2}$ and $\frac{b}{3}$

**c** $\frac{1}{18h}$ and $\frac{1}{9h}$

**d** $\frac{1}{4h}$ and $\frac{1}{4}$

**e** $\frac{7}{15x^4}$ and $\frac{y}{10x^2}$

**f** $\frac{1}{x+4}$ and $\frac{1}{x}$

**g** $\frac{11}{3s}$ and $\frac{17}{5s+54}$

**h** $\frac{9}{x-7}$ and $\frac{9}{x+7}$

**i** $\frac{s}{s+3}$ and $\frac{7}{4(s+3)}$

**j** $\frac{4}{y-8}$ and $\frac{10y}{8-y}$

**k** $\frac{1}{q(q+5)}$ and $\frac{1}{q(q-2)}$

**l** $\frac{8}{p+3}$ and $\frac{8}{(p+3)^2}$

**m** $\frac{1}{(y-7)(y+6)}$ and $\frac{1}{(y-7)(y-4)}$

**n** $\frac{1}{(b+4)(b-8)}$ and $\frac{1}{(b-3)(b-8)}$

**2** Consider the following expression:

$$\frac{3}{x(x+6)} + \frac{10}{(x+6)(x-2)}$$

Find the least common denominator in factored form.

**3** Consider the pair of rational expressions:

$$\frac{6}{(m+3)(m+9)} \quad \text{and} \quad \frac{18+4m}{m^2+12m+27}$$

**a** Factor $m^2+12m+27$.

**b** Now, find the least common denominator in factored form.

**4** Consider the pair of rational expressions:

$$\frac{15+3x}{x+6} \quad \text{and} \quad \frac{11}{x^2+11x+30}$$

**a** Factor $x^2+11x+30$.

**b** Now, find the least common denominator in factored form.

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**5** Consider the pair of rational expressions:

$$\frac{17+2x}{3x^2-108} \quad \text{and} \quad \frac{9}{x^2-36}$$

**a** Factor $x^2-36$.

**b** Factor $3x^2-108$.

**c** Now, find the least common denominator in factored form.

**6** Consider the pair of rational expressions:

$$\frac{3+4y}{y^2+13y+40} \quad \text{and} \quad \frac{15}{y^2+10y+16}$$

**a** Factor $y^2+13y+40$.

**b** Factor $y^2+10y+16$.

**c** Now, find the least common denominator in factored form.

**7** Consider the pair of rational expressions:

$$\frac{17}{6t^2+30t+36} \quad \text{and} \quad \frac{14+3t}{t^2+5t+6}$$

**a** Factor $t^2+5t+6$.

**b** Factor $6t^2+30t+36$.

**c** Now, find the least common denominator in factored form.

## Add and subtract rational expressions

**8** Consider $\frac{2m}{5}$ and $\frac{4m}{6}$.

**a** Find the least common denominator of the two fractions.

**b** Now, simplify $\frac{2m}{5}+\frac{4m}{6}$.

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**9** Fully simplify the following expressions, leaving your answers in factored form:

a $\dfrac{3x}{10}+\dfrac{5x+4}{10}$

b $\dfrac{x}{5}+\dfrac{4x}{7}$

c $\dfrac{11x}{14}+\dfrac{7x}{21}$

d $\dfrac{7x}{11}+\dfrac{3x-2}{44}$

e $\dfrac{8x^2}{11}+\dfrac{5x^2-2x}{55}$

f $\dfrac{x+4}{25}+\dfrac{3x+4}{10}$

g $\dfrac{3x+5}{5}+\dfrac{5x+4}{3}$

h $\dfrac{1}{5}+\dfrac{7}{x-6}$

i $\dfrac{x^2+144}{2\left(x+12\right)}+\dfrac{12x}{x+12}$

j $\dfrac{x+4}{9\left(x+9\right)}+\dfrac{x+4}{x\left(x+9\right)}$

**10** Fully simplify the following expressions, leaving your answers in factored form:

a $\dfrac{6x}{2}-\dfrac{7x}{2}$

b $\dfrac{x+4}{x}+\dfrac{x+3}{x}$

c $\dfrac{3x}{5}-\dfrac{x}{7}$

d $\dfrac{4x}{45}-\dfrac{x}{18}$

e $\dfrac{4x}{7}-\dfrac{2x-3}{21}$

f $\dfrac{3x}{10}-\dfrac{2x+3}{14}$

g $\dfrac{5y}{6}-\dfrac{3y+7}{10}$

h $-3x-\dfrac{3x}{8}$

i $\dfrac{5}{8p}-\dfrac{7}{p}$

j $\dfrac{5x}{7\left(x+1\right)}-\dfrac{10}{x+1}$

k $\dfrac{x^2}{x-4}-\dfrac{16}{x-4}$

l $\dfrac{1}{x\left(x-2\right)}-\dfrac{5}{\left(x^2+6\right)\left(x-2\right)}$

**11** Fully simplify the following expressions, leaving your answers in factored form:

a $\dfrac{5}{8y-5}+\dfrac{10}{7y+7}$

b $\dfrac{5x-y}{x-y}+\dfrac{x-6y}{y-x}$

c $\dfrac{2}{\left(x+3\right)\left(x-10\right)}+\dfrac{5}{\left(x+3\right)\left(x+5\right)}$

d $\dfrac{8}{1+w+z}+\dfrac{7}{9w+9w^2+9wz}$

e $\dfrac{x+8}{x^2+19x+88}+\dfrac{x-7}{\left(x+11\right)\left(x+7\right)}$

f $\dfrac{3}{8x+32}+\dfrac{2}{x^2+4x}$

g $\dfrac{7}{x^2+6x+9}+\dfrac{4}{5x+15}$

h $\dfrac{4}{x^2-9x+20}+\dfrac{5}{x^2+4x-32}$

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**12** Fully simplify the following expressions, leaving your answers in factored form:

**a** $\dfrac{x+6}{x+8}-\dfrac{x-8}{x-6}$

**b** $\dfrac{x}{2x-6}-\dfrac{x+5}{3x+21}$

**c** $\dfrac{x}{x^2-16}-\dfrac{12}{x+4}$

**d** $\dfrac{9}{x-4}-\dfrac{5}{x^2-8x+16}$

**e** $\dfrac{3x}{x^2-25}-\dfrac{2}{(x+3)(x+5)}$

**f** $\dfrac{4}{x^2-15x+54}-\dfrac{3}{x^2-36}$

**g** $\dfrac{5}{xy-3x+4y-12}-\dfrac{7}{11x+44}$

**h** $\dfrac{5}{y-2}-\dfrac{9}{xy-2x-7y+14}$

**13** Determine whether each of the following is equivalent to $3-\dfrac{x-2}{x+3}$:

**a** $2-\dfrac{1}{x+3}$

**b** $2+\dfrac{5}{x+3}$

**c** $2+\dfrac{1}{x+3}$

**d** $2-\dfrac{5}{x+3}$

**14** Determine whether each of the following is equivalent to $\dfrac{25x^2-2}{5x+1}$:

**a** $5x-1+\dfrac{1}{5x+1}$

**b** $5x-1-\dfrac{1}{5x+1}$

**c** $5x+1+\dfrac{1}{5x+1}$

**d** $5x+1-\dfrac{1}{5x+1}$

## Application

**15** In an upcoming election, it is anticipated that the number of men who vote, $x$, will be greater than the number of women who vote, $y$. Each expression below represents the expected number of people who will not vote in two different counties:

- **County A:** $\dfrac{x-y}{2}$
- **County B:** $x-\dfrac{y}{2}$

Which county expects more people to not vote?

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